

TOOWOOMBA, Australia

WMO: 955510

Lat: 27.5425S

Lon: 151.9133E

Elev: 642

StdP: 93.85

Time Zone: 10.00 (E10)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 3.5	(c) 4.9	(d) -6.7	(e) 2.3	(f) 15.4	(g) -4.6	(h) 2.8	(i) 13.5	(j) 12.0	(k) 11.7	(l) 11.0	(m) 11.0	(n) 4.6	(o) 240	(p) 0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 9.3	(c) 32.4	(d) 18.9	(e) 30.7	(f) 18.9	(g) 29.1	(h) 18.7	(i) 22.1	(j) 27.0	(k) 21.5	(l) 26.1	(m) 20.8	(n) 25.3	(o) 5.5	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	16.8	23.6	20.1	16.0	23.0	19.5	15.4	22.4	68.2	27.0	65.6	26.1	63.2	25.2	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.9	(b) 10.6	(c) 9.8	(e) 0.5	(f) 35.6	(g) 1.2	(h) 1.9	(i) -0.4	(j) 37.0	(k) -1.1	(l) 38.2	(m) -1.7	(n) 39.2	(o) -2.6	(p) 40.7		
			DB													
			WB													
				(e) -0.8	(f) 23.4	(g) 0.9	(h) 1.1	(i) -1.4	(j) 24.2	(k) -1.9	(l) 24.8	(m) -2.4	(n) 25.5	(o) -3.0	(p) 26.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	22.7	22.0	20.8	18.0	14.7	12.1	11.6	12.9	16.2	18.4	20.2	21.7
	DBStd	4.63	2.35	2.45	2.07	2.07	2.39	2.55	2.47	2.69	2.98	2.96	2.83	2.75
	HDD10.0	28	0	0	0	0	2	10	12	5	0	0	0	0
	HDD18.3	838	1	1	2	29	113	187	208	169	78	35	12	4
	CDD10.0	2803	393	337	336	241	148	73	62	96	186	261	307	363
	CDD18.3	571	135	105	79	19	2	0	0	2	13	38	69	109
	CDH23.3	3574	898	608	354	62	6	1	0	12	109	289	486	749
CDH26.7	1042	290	186	76	3	1	0	0	3	19	72	147	246	
Wind	WSAvg	5.6	6.4	6.6	6.5	5.6	4.9	4.9	4.7	5.0	5.3	5.7	6.0	6.1
Precipitation	PrecAvg	937	122	116	78	58	68	47	48	40	39	77	100	127
	PrecMax	1479	520	421	174	421	520	373	188	188	110	234	303	271
	PrecMin	469	17	11	2	0	6	2	0	0	3	8	15	11
	PrecStd	233	87	86	51	72	84	62	43	33	30	48	62	63
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.3	32.0	27.2	24.0	20.5	20.9	24.7	29.0	31.6	32.7	34.3
		MCWB	20.3	20.4	18.7	16.9	15.2	13.5	12.6	13.5	15.3	16.0	17.3	18.2
	2%	DB	31.6	31.1	28.8	25.1	21.8	18.9	19.0	21.7	26.2	28.6	30.4	31.7
		MCWB	19.7	20.2	19.3	16.8	14.3	13.0	11.9	12.1	14.2	15.6	17.4	18.7
	5%	DB	29.8	28.8	26.8	23.7	20.4	17.6	17.5	19.8	24.2	26.5	28.2	29.3
		MCWB	19.7	20.0	18.8	16.3	13.6	12.4	11.4	11.5	13.8	15.3	17.3	18.7
	10%	DB	27.9	26.9	25.2	22.3	19.2	16.4	16.2	18.0	22.2	24.5	26.1	27.3
		MCWB	19.6	19.5	18.4	15.9	13.2	12.0	11.0	10.9	13.3	15.0	17.1	18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.0	23.3	22.0	19.5	17.6	15.9	16.1	15.7	17.5	19.7	21.1	22.2
		MCDB	28.4	29.2	25.7	23.6	19.7	18.0	18.1	20.8	22.5	25.1	27.4	28.4
	2%	WB	22.2	22.1	21.1	18.4	16.3	14.8	14.6	14.4	16.5	18.3	20.0	21.2
		MCDB	27.0	26.6	24.6	22.0	19.0	16.9	16.4	18.3	22.0	23.1	24.9	26.4
	5%	WB	21.5	21.2	20.3	17.7	15.5	14.0	13.3	13.3	15.5	17.4	19.2	20.5
		MCDB	26.1	25.9	23.5	21.0	18.2	16.0	15.5	17.1	20.9	22.1	24.1	25.3
	10%	WB	20.8	20.4	19.6	17.0	14.7	13.2	12.1	12.3	14.7	16.6	18.4	19.9
		MCDB	25.2	25.0	22.9	20.3	17.5	15.2	14.7	16.1	19.5	21.3	23.1	24.3
Mean Daily Temperature Range	5% DB	MDBR	9.3	8.7	8.4	8.5	8.6	8.0	8.8	9.8	10.5	10.4	10.1	9.6
		MCDBR	12.6	11.7	10.8	10.1	9.8	8.9	9.9	11.4	12.9	13.6	13.3	12.8
		MCWBR	3.5	3.4	3.3	3.5	4.2	4.4	4.8	5.0	4.4	4.5	4.0	3.9
	5% WB	MCDBR	9.7	9.7	8.3	8.0	7.2	6.5	7.1	9.0	10.6	10.4	10.5	10.1
		MCWBR	3.8	3.7	3.4	3.4	3.5	3.9	4.5	4.8	4.4	4.4	4.1	4.0
Clear-Sky Solar Irradiance	taub	0.351	0.343	0.324	0.307	0.288	0.282	0.275	0.285	0.318	0.339	0.349	0.358	
	taud	2.552	2.587	2.642	2.660	2.668	2.688	2.695	2.636	2.527	2.498	2.510	2.509	
	Ebn at Noon	991	983	973	944	922	907	929	954	962	975	986	985	
	Edh at Noon	109	103	92	82	74	69	71	83	102	111	113	115	
All-Sky Solar Radiation	RadAvg	6.78	6.20	5.39	4.68	3.85	3.19	3.64	4.59	5.62	6.27	6.80	6.81	
	RadStd	0.54	0.57	0.33	0.22	0.20	0.25	0.28	0.34	0.44	0.60	0.64	0.68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.41	+0.52	N/A	N/A	N/A	N/A	N/A	-67	+186	+112

Nomenclature: See separate page